

Interest Rate and Credit Models

Homework Assignment #1

Andrew Lesniewski
Baruch College, CUNY

February, 2026

Research Project: Factor Models of Interest Rate Curves

This is an open-ended research project on *factor models for the yield curve*, with an emphasis on (i) curve fitting (cross-sectional), (ii) factor dynamics (time-series), and (iii) applications such as forecasting or trading.

Core model: *Nelson–Siegel (NS)*. Let $y(s)$ denote the (instantaneous) forward rate curve at a fixed observation date (denoted by $f_0(s)$ in the lecture notes). In the Nelson–Siegel model, $y(s)$ is assumed to have parametric form

$$y(s) = \beta_0 + \beta_1 e^{-\lambda s} + \beta_2 \lambda s e^{-\lambda s}, \quad \beta_0, \beta_1, \beta_2 \in \mathbb{R}, \lambda > 0. \quad (1)$$

The corresponding zero-coupon yield $Y(T)$ is defined by

$$Y(T) = \frac{1}{T} \int_0^T y(s) ds. \quad (2)$$

Parts (A) - (C) below constitute the *core required component* of the project. Parts (D) - (E) are extensions and may be approached selectively.

(A) *Closed form yields.* Show that

$$Y(T) = \beta_0 + \beta_1 \frac{1 - e^{-\lambda T}}{\lambda T} + \beta_2 \left(\frac{1 - e^{-\lambda T}}{\lambda T} - e^{-\lambda T} \right). \quad (3)$$

Interpret $\beta_0, \beta_1, \beta_2$ economically (“level / slope / curvature” heuristics).

(B) *Data construction (daily, since Jan 1, 2016).* Build a daily time series of zero-coupon yields at maturities

$$T \in \{0.25, 0.5, 1, 2, 3, 5, 7, 10\} \text{ years}$$

(you may extend beyond 10Y if you want). Acceptable sources include:

- (i) *Zero-coupon curve estimates* from the Federal Reserve Board / Gürkaynak-Sack-Wright dataset (or a closely related release mirrored on FRED).¹
- (ii) Bloomberg or WRDS fixed income data.
- (iii) Alternatively (and preferably): *swap* zero rates at the same maturities.

Important: If you use Treasury par yields (e.g. constant-maturity yields), briefly explain the distinction and how it may bias the NS fit.

(C) *Cross-sectional calibration (curve fitting).* For each day t , let $\{Y_t(T_i)\}_{i=1}^n$ be the observed yields. Formulate NS calibration as a nonlinear least squares problem:

$$\min_{\beta_0, \beta_1, \beta_2, \lambda} \sum_{i=1}^n w_i \left(Y_t(T_i) - \hat{Y}(T_i; \beta_0, \beta_1, \beta_2, \lambda) \right)^2,$$

with weights w_i of your choice (equal weights is fine; justify other choices).

- (i) Implement the calibration and report fitting error on several representative days (e.g. calm regime vs. stressed regime). Include plots of observed vs. fitted curves.
- (ii) *Choosing λ .* Because λ is often unstable, compare at least two of the following approaches:
 - (a) grid search over λ with linear least squares for $(\beta_0, \beta_1, \beta_2)$ conditional on λ ,

¹This avoids the “par yield” issue.

- (b) fixing λ at a constant value using a rule or cross-validation objective you define,
- (c) full nonlinear optimization over all parameters.

Quantify how your λ choice affects fit quality and parameter stability.

- (iii) Study the *time series stability* of $(\beta_0, \beta_1, \beta_2)$: summary statistics, rolling volatility, correlations, and (optionally) regime dependence.

(D) **Dynamic factor curve models (extension).** Move beyond “fit each day independently” by imposing a time-series structure on the factors. Choose one of the following directions (or propose your own):

- (D1) *Dynamic Nelson-Siegel (DNS)*. Treat $(\beta_{0,t}, \beta_{1,t}, \beta_{2,t})$ as latent states following a VAR(1) (or AR(1) componentwise), and estimate them using a *state-space / Kalman filter* framework. Compare filtered and smoothed factors to the two-step daily OLS / nonlinear fits.
- (D2) *Nelson-Siegel-Svensson (NSS)*. Extend the curve with an additional curvature term and evaluate whether the improved cross-sectional fit is worth the increased estimation variance. (Optional: combine with (D1) in state space form.)

Evaluate models via out-of-sample curve fit and/or forecasting accuracy using a rolling-window design.

(E) **Applications: forecasting and / or trading (extension).** Design a systematic study using your fitted factors or yields. This component is intentionally open-ended, but you are encouraged to focus on *curve trades* rather than outright duration bets. Examples include:

- (i) forecasting yields or yield changes at selected maturities,
- (ii) trading canonical curve strategies such as:
 - (a) DV01-neutral steepeners (e.g. 2s10s),
 - (b) DV01-neutral flatteners,
 - (c) DV01-neutral butterfly trades (e.g. 2s5s10s),
- (iii) comparing linear time series models (AR / VAR) with ML approaches (XGBoost, LightGBM, neural networks, transformers),
- (iv) incorporating macroeconomic or market variables as additional predictors (optional).

If you backtest, include transaction cost assumptions, risk limits, and at least one benchmark strategy.

Deliverables.

- A short report (6-10 pages, plus appendix if needed) with figures and clear methodology.
- Reproducible code (notebook or scripts) and a brief “how to run” note.
- A 15-20 minute presentation. **Presentations will take place on March 18.**

Suggested references. I have enclosed several papers that may be helpful. Some of these references are classic contributions to the theory and empirical implementation of factor models for the yield curve, while others are more recent. You are encouraged to consult these papers for background, intuition, and methodological ideas, but you are not expected to replicate any specific model.

You are welcome to search for and include additional references that are relevant to your project. The use of AI tools for idea shaping, literature exploration, and code development is strongly encouraged (with proper attribution).